

Impurity Metrics for Continuous Attributes

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Prerequisites

Before proceeding with this chapter, students must have completed:

- Decision Tree Level 1
- Decision Tree Level 2 [Impurity Metrics, the earlier notebook]

Learning Objectives

After going through this notebook, students should be able to:

- Find the best split when the dataset has continuous attributes

In the previous notebook, we discussed impurity metrics and demonstrated how we compute them for discrete attributes. In this notebook, we will see how we can compute impurity metrics for continuous attributes. We will use gini as the impurity metric, but the process can be used to compute other impurity metrics. We will use the following dataset for the demonstration:

Dataset

Yearly Income(in thousand dollars)	Net Worth(in million dollars)	Will repay loan
120	2.5	yes
250	1.0	yes
130	1.0	no
80	3.0	yes

We will use the attribute **Net Worth** to compute the gini index. The step involved in the process are:

1. Arrange all attribute values in ascending order. Ignore duplicates.

Sorted attribute values = [1.0, 2.5, 3.0]

Sorting values is important because if values are not sorted, then we may not be able to explore all possible threshold values and hence the splits. The resulting best split obtained in such a way may not be the most optimal split.

2. Find the average of two consecutive numbers in the sorted attribute values list. These are used as threshold values.

averaged attribute values = [1.75, 2.75]

3. Use each of these threshold values as a split value and compute gini for child nodes. All instances with attribute value less than or equal to the threshold value form the left child and the rest form the right child.

Example: Using 1.75 as the split value, we get following splits:

Split 1: Net Worth <= 1.75

Yearly Income(in thousand dollars)	Net Worth(in million dollars)	Will repay loan
250	1.0	yes
130	1.0	no

Split 2: Net Worth > 1.75

Yearly Income(in thousand dollars)	Net Worth(in million dollars)	Will repay loan
120	2.5	yes
80	3.0	yes

Gini for **Split 1, Net Worth** ≤ 1.75, can be computed as:

$$\begin{aligned}
 Gini_{split\ 1} &= 1 - [P_{\text{Will repay loan} = \text{yes}}^2 + P_{\text{Will repay loan} = \text{no}}^2] \\
 &= 1 - [(\frac{1}{2})^2 + (\frac{1}{2})^2] \\
 &= 0.5
 \end{aligned}$$

Gini for **Split 2, Net Worth** > 1.75, can be computed as:

$$\begin{aligned}
 Gini_{split\ 2} &= 1 - [P_{\text{Will repay loan} = \text{yes}}^2 + P_{\text{Will repay loan} = \text{no}}^2] \\
 &= 1 - [(\frac{2}{2})^2 + 0] \\
 &= 0
 \end{aligned}$$

Gini for the attribute **Net Worth** can be computed by taking a weighted average of $Gini_{split\ 1}$ and $Gini_{split\ 2}$.

$$\begin{aligned}
 Gini_{\text{Net Worth}} &= \frac{2}{4} * Gini_{split\ 1} + \frac{2}{4} * Gini_{split\ 2} \\
 &= \frac{2}{4} * 0.5 + \frac{2}{4} * 0 \\
 &= 0.25
 \end{aligned}$$

4. Repeat step 3 for each threshold values of each attribute.

Attribute	Threshold Value	Gini
Net Worth	1.75	0.25
Net Worth	2.75	0.33
Yearly Income	100	0.33
Yearly Income	125	0.25
Yearly Income	190	0.33

Figure 1: Attribute, threshold value, and their corresponding gini

5. Attribute and threshold value that correspond to the minimum gini is used to make the split. In case of a tie, select one randomly.

Attribute	Threshold Value	Gini
Net Worth	1.75	0.25
Net Worth	2.75	0.33
Yearly Income	100	0.33
Yearly Income	125	0.25
Yearly Income	190	0.33

Figure 2: Entries highlighted in green correspond to minimum gini

In our case, both **Net Worth** at threshold value 1.75 and **Yearly Income** at threshold value 125 yields minimum gini. Anyone of them can be selected at random to make the split.

In this way, we can find the best split for continuous attributes.

In the next chapter, we will discuss decision tree algorithms. We will discuss in brief about impurity metric and pruning method used by each of those algorithms.

Key Takeaways

- To find the best split for a continuous attribute, we need to compute gini(or any other impurity metric of choice) for each threshold value of the attribute
- If we have multiple candidates for the best split(minimum gini), we can select any of those to create the split.

Additional Resources

Split Value for Continuous Attribute

- [Choosing Split value for continuous attribute](https://datascience.stackexchange.com/questions/24339/how-is-a-splitting-point-chosen-for-continuous-variables-in-decision-trees)
(<https://datascience.stackexchange.com/questions/24339/how-is-a-splitting-point-chosen-for-continuous-variables-in-decision-trees>) Refer this stack exchange question to find out how we can select split points for continuous attributes